# 2.3.7. Instrument control for linear calibration

*Purpose*

The purpose of the control program is to guarantee that the calibration of an instrument does not degrade over time.

*Approach*

This is accomplished by exercising quality control on the instrument's output in much the same way that quality control is exercised on components in a process using a modification of the Shewhart control chart.

*Check standards needed for the control program*

For linear calibration, it is sufficient to control the end-points and the middle of the calibration interval to ensure that the instrument does not drift out of calibration. Therefore, check standards are required at three points; namely,

- at the lower-end of the regime
- at the mid-range of the regime
- at the upper-end of the regime

*Data collection*

One measurement is needed on each check standard for each checking period. It is advisable to start by making control measurements at the start of each day or as often as experience dictates. The time between checks can be lengthened if the instrument continues to stay in control.

*Definition of control value*

To conform to the notation in the section on instrument corrections, $X^*$ denotes the known value of a standard, and $X$ denotes the measurement on the standard.

A control value is defined as the difference

$$W = X^* - X$$

If the calibration is perfect, control values will be randomly distributed about zero and fall within appropriate upper and lower limits on a control chart.

*Calculation of control limits*

The upper and lower control limits (Croarkin and Varner)) are, respectively,

$$l_{upper} = +\frac{s}{\hat{b}} t^*_{\alpha/2}(\nu)$$

$$l_{lower} = -\frac{s}{\hat{b}} t^*_{\alpha/2}(\nu)$$

where $s$ is the residual standard deviation of the fit from the calibration experiment, and $\hat{b}$ is the estimated slope of the linear calibration curve.

*Values t**

The critical value, $t^*_{\alpha/2}$, can be found in the *t*table*; $\nu$ is the degrees of freedom for the residual standard deviation; and $\alpha$ is equal to 0.05.

*Determining t**

For the case where $\alpha = 0.05$ and $\nu = 38$, the critical value of the $t^*$ statistic is 2.497575.

R code and Dataplot code can be used to determine $t^*$ critical values using a standard $t$-table for the $\zeta$ quantile and $\nu$ degrees of freedom where $\zeta$ is computed as

$$\zeta = \frac{1}{2}\left[1 - e^{\frac{\ln(1-\alpha)}{m}}\right]$$

where $m$ is the number of check standards.

*Sensitivity to departure from linearity*

If

$$l_{lower} \leq W \leq l_{upper}$$

the instrument is in statistical control. Statistical control in this context implies not only that measurements are repeatable within certain limits but also that instrument response remains linear. The test is sensitive to departures from linearity.

*Control chart for a system corrected by a linear calibration curve*

An example of measurements of line widths on photomask standards, made with an optical imaging system and corrected by a linear calibration curve, are shown as an example. The three control measurements were made on reference standards with values at the lower, mid-point, and upper end of the calibration interval.



Site Privacy | Accessibility | Privacy Program | Copyrights | Vulnerability Disclosure |

No Fear Act Policy | FOIA | Environmental Policy | Scientific Integrity | Information Quality Standards |

Commerce.gov | Science.gov | USA.gov | Vote.gov

NIST
National Institute of Standards and Technology
U.S. Department of Commerce